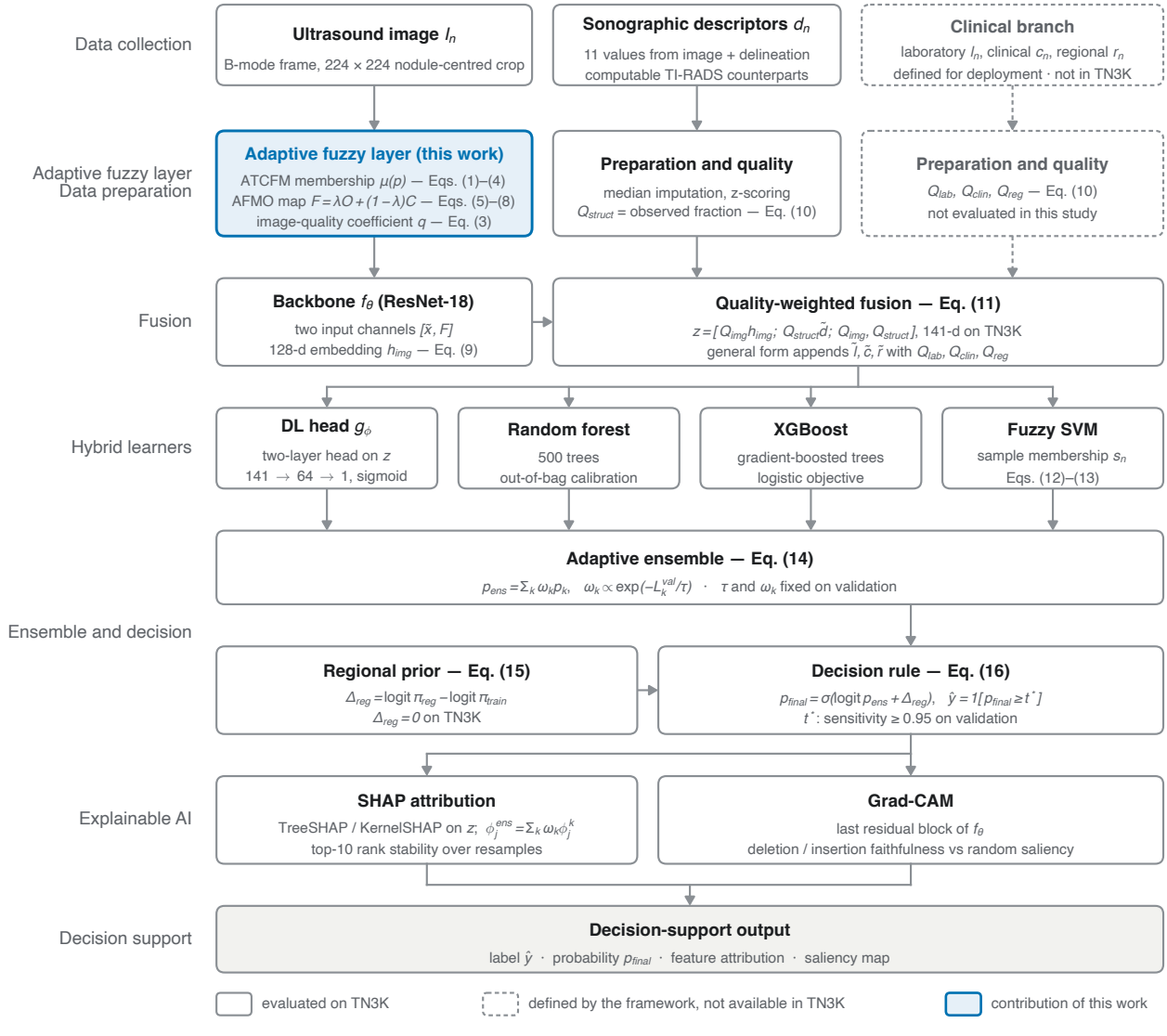

Data collection
Ultrasound image I_n
B-mode frame, 224 × 224 nodule-centred crop
Sonographic descriptors d_n
11 values from image + delineation
computable TI-RADS counterparts
Clinical branch
laboratory l_n, clinical c_n, regional r_n
defined for deployment · not in TN3K
Adaptive fuzzy layer
Data preparation
Adaptive fuzzy layer (this work)
ATCFM membership $\mu(p)$ — Eqs. (1)–(4)
AFMO map $F=\lambda O+(1-\lambda)C$ — Eqs. (5)–(8)
image-quality coefficient q — Eq. (3)
Preparation and quality
median imputation, z-scoring
Q_{struct} = observed fraction — Eq. (10)
Preparation and quality
Q_{lab}, Q_{clin}, Q_{reg} — Eq. (10)
not evaluated in this study
Fusion
Backbone f_θ (ResNet-18)
two input channels $[\tilde{x}, F]$
128-d embedding h_{img} — Eq. (9)
Quality-weighted fusion — Eq. (11)
$z=[Q_{img}h_{img};\ Q_{struct}\tilde{d};\ Q_{img},\ Q_{struct}]$, 141-d on TN3K
general form appends $\tilde{l}$, $\tilde{c}$, $\tilde{r}$ with Q_{lab}, Q_{clin}, Q_{reg}
Hybrid learners
DL head g_ϕ
two-layer head on z
141 → 64 → 1, sigmoid
Random forest
500 trees
out-of-bag calibration
XGBoost
gradient-boosted trees
logistic objective
Fuzzy SVM
sample membership s_n
Eqs. (12)–(13)
Ensemble and decision
Adaptive ensemble — Eq. (14)
$p_{ens}=\Sigma_k \omega_k p_k$, $\omega_k \propto \exp(-L_k^{val}/\tau)$ · τ and ω_k fixed on validation
Regional prior — Eq. (15)
$\Delta_{reg}=\mathrm{logit}\,\pi_{reg}-\mathrm{logit}\,\pi_{train}$
$\Delta_{reg}=0$ on TN3K
Decision rule — Eq. (16)
$p_{final}=\sigma(\mathrm{logit}\,p_{ens}+\Delta_{reg})$, $\hat{y}=1[p_{final}\geq t^*]$
t^*: sensitivity ≥ 0.95 on validation
Explainable AI
SHAP attribution
TreeSHAP / KernelSHAP on z; $\phi_j^{ens}=\Sigma_k \omega_k \phi_j^k$
top-10 rank stability over resamples
Grad-CAM
last residual block of f_θ
deletion / insertion faithfulness vs random saliency
Decision support
Decision-support output
label $\hat{y}$ · probability p_{final} · feature attribution · saliency map
evaluated on TN3K
defined by the framework, not available in TN3K
contribution of this work